

Navigating the Fog of War:



**A Programmatic Approach to Capturing and Communicating
Relevant Insights from Rising Geopolitical Tensions**

Agenda



- Expectations
- Background Context
- Case Study 1: Russia's Invasion of Ukraine
- Case Study 2: Iran-Hamas-Israel
- Case Study 3: China-Taiwan

Expectations

This talk will not...

- Likely win hearts and minds
- Provide a get out of jail free card for shoddy analysis
- Make you an expert on international relations or multi-order implications to cyber security for current events

This talk is designed to...

- Teach you something new
- Provide CTI teams with a starting point to save finite time and energy
- Apply a formulaic approach to determining relevance
- Assist in avoiding shiny object syndrome
- Deconstruct the dynamics of escalating geopolitical tensions
- Identify impactful storytelling approaches and timing to various stakeholders

Background Context

WHAT WOULD YOU SAY...

YOU DO HERE...

CTI provides timely and relevant insights about the cyber threat landscape, cyber threat actors, their capabilities, and motivations to inform risk exposure, planning, or cyber defense actions.

Background Context

Today's geopolitical environment is increasingly complex. PESTLE and DIMEFIL are models to compare elements of national power. Cyber, as a tool of statecraft, is interwoven across multiple DIMEFIL categories to aid countries in achieving their national-level objectives.

PESTLE

Political	Economic
• Government Policies • Political Stability • Trade Regulation	• Economic impact • Inflation / growth • Changes in the economic environment
Social	Technological
• Socio-cultural Change • Ethnic / Religion • Opinions • Living Standards	• Emerging Technologies & R&D • Innovation • Automation
Environmental	Legal
• Perception • Climate Change • Sustainability • Resources	• Data Privacy & Protection • Regulations (NIS, DORA in Europe for e.g)

DIMEFIL

	Diplomatic	Information
	• Treatises • Bi/multi-lateral Partnerships • International Forums	• Public Diplomacy • Overt and Covert Messaging
	Military	Economic
	• Force Comp • Security Cooperation and Capacity Building • Show of Force	• Foreign Aid • Trade Agreements • Tariffs • Embargos • Sanctions
Financial	Intelligence	Law Enforcement
• Targeted Sanctions • Freeze Assets	• Information Sharing • Partnerships • Collection Operations	• Legal Frameworks • International Partners • Investigation and Prosecution

Background Context

High level threat actor use cases:

				
Gain Insights • Espionage • Counterespionage	Profit Financially • Economic Espionage • Extortion • Fraud • Theft	Shape Behavior • Cyber-Enabled Information Operations • Demonstration of Capability • Exposure • Targeted Attacks	Prepare for War • Operational Preparation	Support War Efforts • Standalone Operations • Force Multiplier Operations

Background Context

For nations, cyber is an asymmetric tool used as a form of soft power to augment existing tools of statecraft. Cyber operations can be used for a variety of purposes:

Espionage

Political

- Provide situational awareness of organizational decisions and plans
- Inform course of action in bi- or multi-lateral negotiations
- Identify potentially compromising information to use as a bartering chip

Military

- Understand capabilities, plans, decision calculus, and key stakeholders

Economic

- Bolster domestic competitiveness
- Fund regime coffers

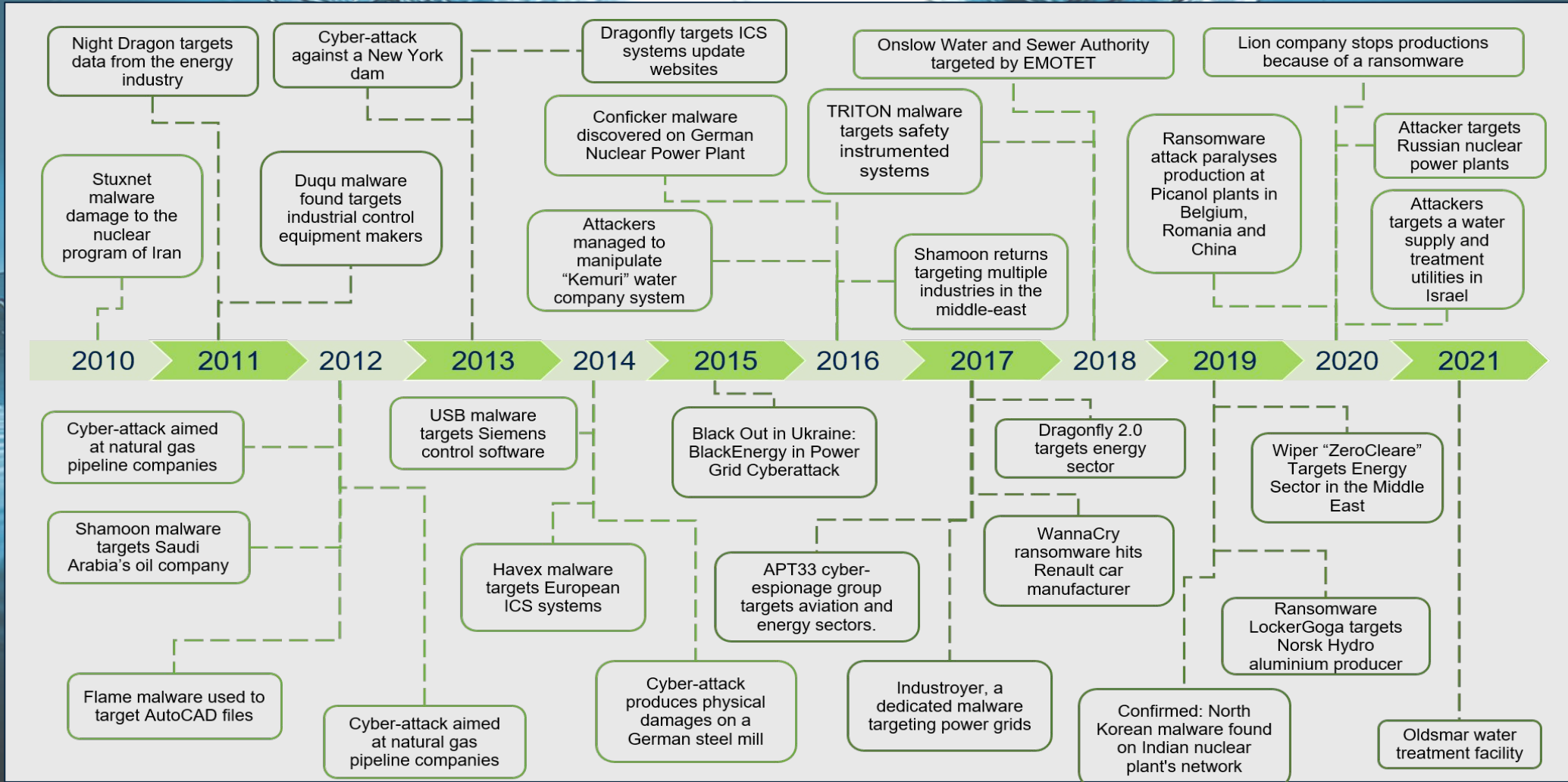
Attack

- Disrupt or degrade the availability of information systems and network access
- Destroy systems storing information
- In support of concurrent or near-term additional operations
- Cause embarrassment
- Extortion
- Misdirection or as a false flags

Background Context

Cyber attacks (“wiper” or otherwise) are leveraged by countries dating back to 2010

Intended targets vs. collateral damage to achieve an effect must be considered



Background Context

Cyber “Norms” During Peace Time vs. War Time

Cyber operations calculus – risk tolerance for discovery, attribution, and willingness to take more aggressive actions – shift when we move from “peace time” to “war time”.

- Multiple priority requirements likely predicate a shift frequency, volume, and operational targeting. i.e., what was normal ops tempo can no longer be used as a baseline, some operations will be put into an inactive mode with periodic check ins in order to balance conflicting priorities.
- There may be a willingness to burn existing access to support broader mission objectives, such as to delay an enemy’s military movement or to buy time for a military advance while controlling the public narrative.
- New accesses are not instantaneous, reconnaissance – scanning activity, email address discovery, and phishing – will be required for new targets.
- Alternative tradecraft and tools may be used to obscure linkage to past operations. Expect to see “false flags” more regularly or an application of the YOLO strategy.

Background Context

Geopolitical conflicts sometimes have a cyber component or cyber implications. Cyber as a tool to project power:

Israel, Iran and Cyber Warfare

MAY 26, 2020, 5:31 PM

Just days after the Israeli media reported earlier this month that Iranian military forces were being withdrawn from Syria, Israelis learned that Iran had sprung a nasty surprise on Israel.

Iran, Israel's most formidable enemy, had launched a cyberattack on Israel's water and sewage facilities. Channel 12 broke the news on May 9, but the actual Iranian hack took place on April 24-25.

Iran's cyber breach of Israel's water command and control systems caused only minor damage, but it set off alarm bells in the Israeli government because Iran had crossed a red line by having targeted a critical component of Israel's civil infrastructure.

To no one's surprise, Israel responded forcefully to Iranian aggression. On May 9, Israel launched a cyberattack on the Iranian port of Shahid Rajaei, through which 60 percent of Iran's trade passes to or from the Strait of Hormuz and which houses its most important naval base.

Israel's retaliatory strike, designed to be limited in scope, led to delays in shipments and traffic jams, prompting observers to speculate that the real point of the reprisal was psychological rather than strategic.

Israel's unmistakable message was that its cyber capabilities are among the most advanced in the world and that further Iranian attempts to sabotage its infrastructure would be far more costly to Iran.

Iran gas stations reportedly hit by massive cyberattack

Hackers in Iran address Iran Supreme Leader Ayatollah Ali Khamenei demanding, 'Where is the gas?'

By YONAH JEREMY BOB
OCTOBER 26, 2021 21:41



Gas stations across Iran malfunctioned on Tuesday, reportedly due to a massive cyberattack, according to Iranian state media.

With the details still hazy, speculation is rife about whether the purported attack came from the US, Israel or from local Iranian anti-regime groups.

According to reports, messages were posted in some systems that were [hacked](#), addressing Iran's Supreme Leader Ayatollah Ali Khamenei directly and demanding to know, "Where is the gas?" The attack comes some two years after nationwide protests over gas shortages in fall 2019.

Satellite hack on eve of Ukraine war was a coordinated, multi-pronged assault

The satellite hack that took the world by storm was more complex than initially thought, according to a Viasat executive.

BY CHRISTIAN VASQUEZ AND ELIAS GROLL • AUGUST 10, 2023

LAS VEGAS — The cyberattack that crippled satellite communications on the eve of the Ukraine war was more broad than initially understood and carried out by attackers with detailed knowledge of the compromised system, an executive with Viasat, whose modems were targeted in the attack, revealed during a talk Thursday at the Black Hat cybersecurity conference in Las Vegas.

When hackers attacked Viasat as Russian forces prepared to stream across Ukraine's border, they relied on a piece of malware that wiped the contents of thousands of targeted modems. That component of the attack has been fairly well understood, but on Thursday, Mark Colaluca, vice president and chief information security officer at Viasat Corporate, revealed a second, previously unknown component of the attack and said that the company remains under assault.

That secondary line of attack used "highly technical knowledge of our network" and the networking protocols it relies on to "target specific terminals to not let them back on the network," Colaluca said.

U.S. government and Ukrainian officials have blamed Russia for the attack on the Viasat KA-SAT network that shut down communications that Kyiv relied on for commanding troops and thousands of European used for internet access. And the secondary attack revealed on Thursday showed that once Russian hackers had disrupted internet access in the first place they used sophisticated methods to try and prevent it from being restored.

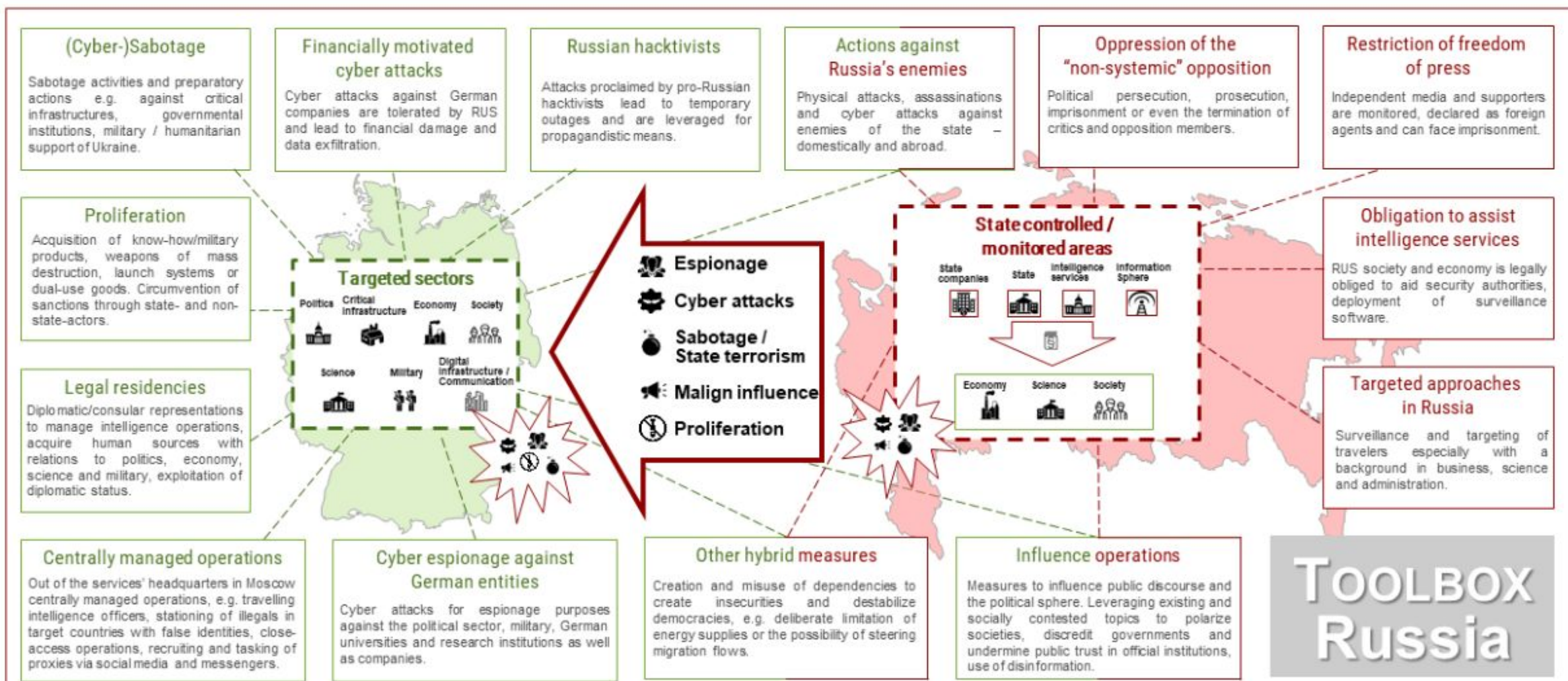


Case Study 1: Russia's Invasion of Ukraine

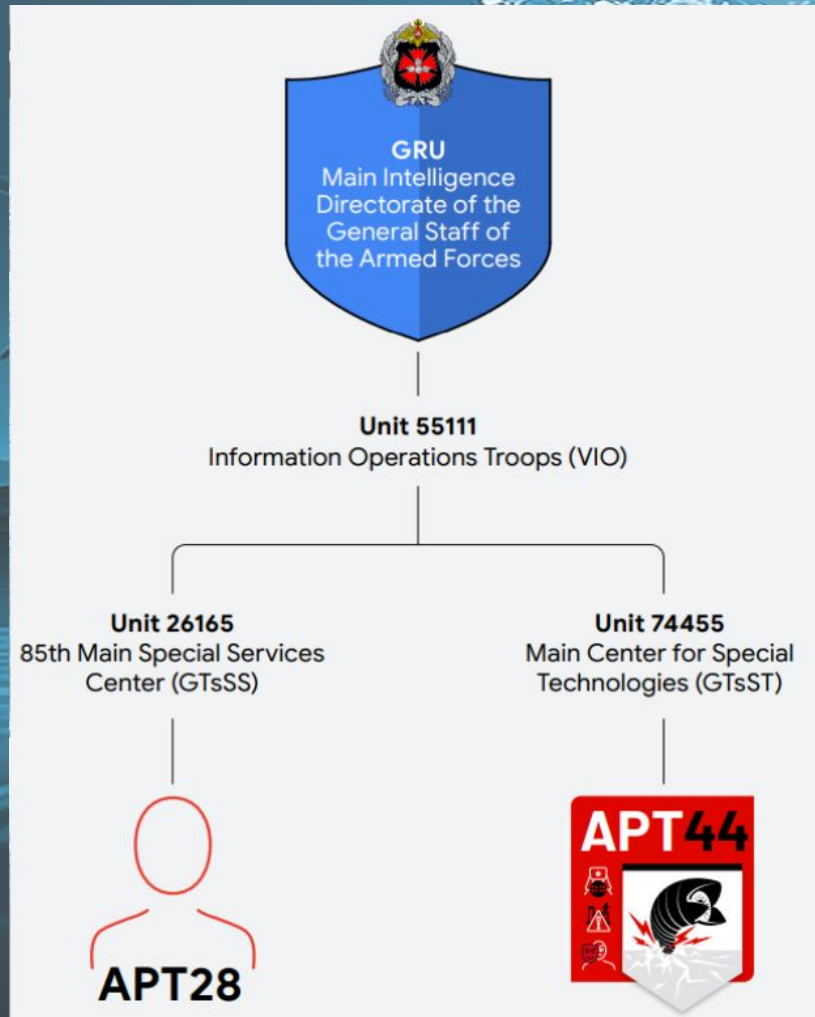
Russia's Operational Methods Against Germany and Western Democracies



Bundesamt für
Verfassungsschutz



Russian Intelligence Support to the Invasion



GRU Unit 29155

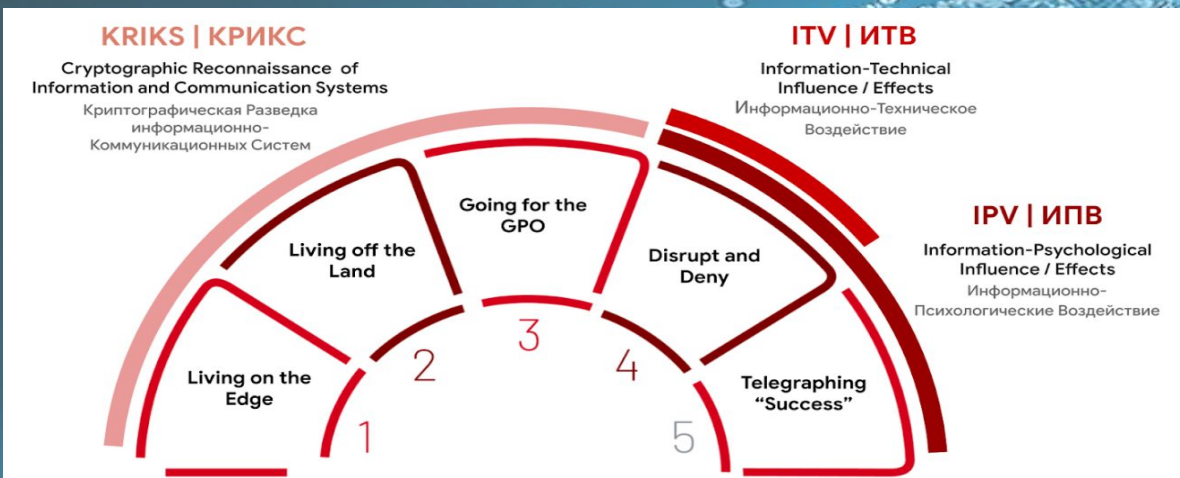
FSB

Turla

UNC530



GRU's Information Confrontation Doctrine



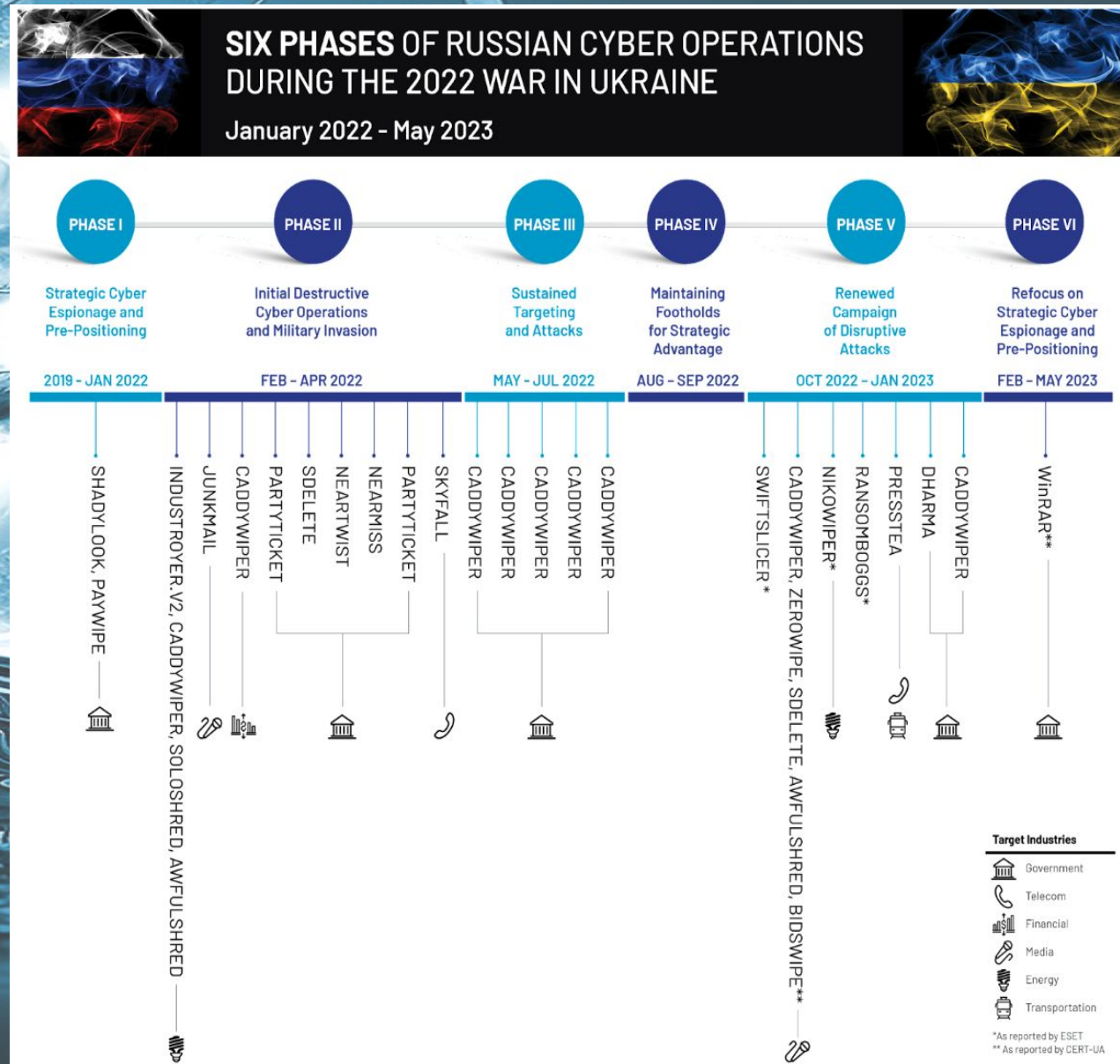
1. Living on the Edge: Leveraging hard-to-detect compromised edge infrastructure such as routers, VPNs, firewalls, and mail servers.

2. Living off the Land: Using built-in tools such as operating system components or pre-installed software.

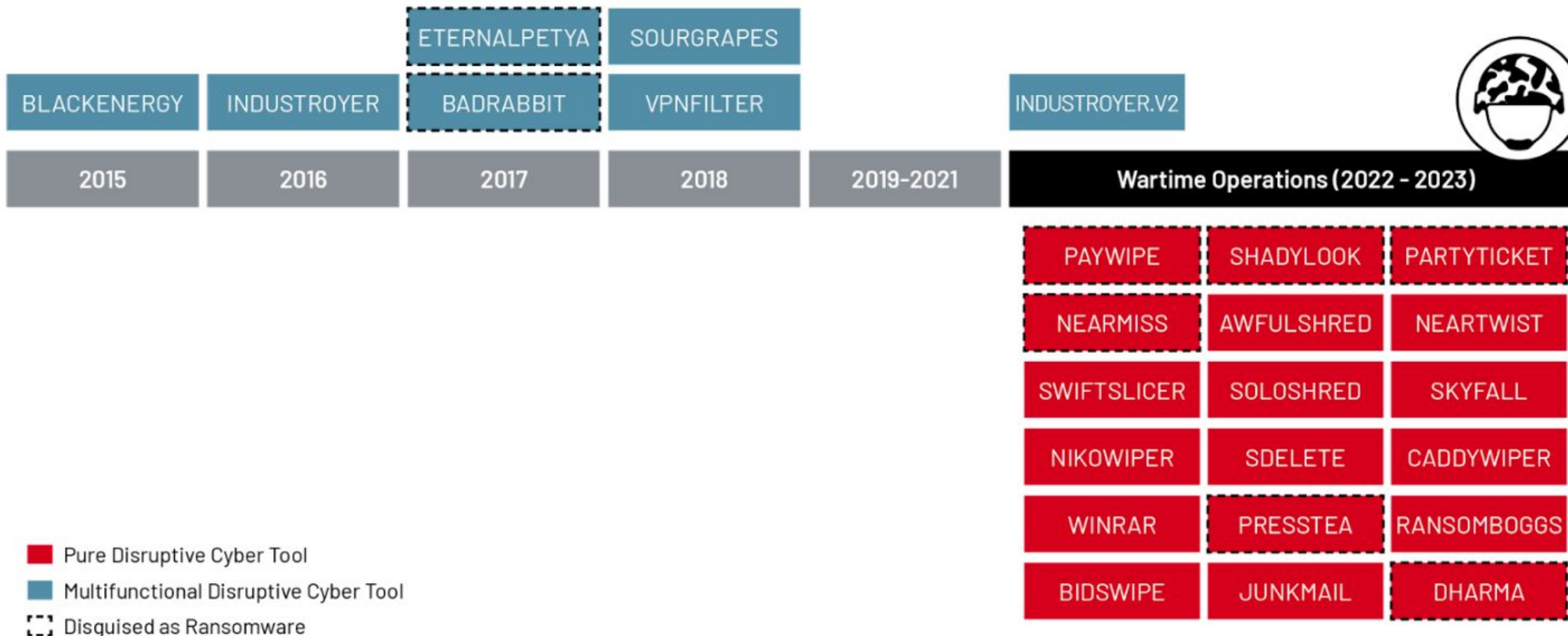
3. Going for the GPO: Creating persistent, privileged access from which wipers can be deployed via group policy objects (GPO) using a tried-and-true PowerShell script.

4. Disrupt and Deny: Deploying "pure" wipers and other low-equity disruptive tools such as ransomware to fit a variety of contexts and scenarios.

5. Telegraphing "Success": Amplifying the narrative of successful via hacktivist personas regardless of the actual impact of the operation.



GRU's Information Confrontation Doctrine



GRU Recruiting, Influencing, and Leveraging Hacktivists

An evolution of a playbook technique they haven't employed since 2007 in their attack against Estonia



Holy League
#OpBelgium
#OpGERM
#OpGermany
#Holy_league
#AllianceNoName0516



**Fighter Blackhat
Cyber Crime**
#AllianceNoName0516



Wolf Cyber Army
#AllianceNoName0516



XSEC 404
#AllianceNoName0516



CyberVolk Anonymous
@YourAnonTV
@AnonOpsSE
@YourAnonCentral
@xVENDETTAMAFIAx
@Noname0516
@YourAnonSurge_
#OpFrance
#OpGeorgia



NoName57(16)
#NoName057
#NoName0516
#OP404
#FUCKMELONI
#OpItaly
#ScioperoGenerale
#OpDenmark
#OpFrance
#Cyber_Islamic_
resistance
#ForRussia



EvilByte
#AllianceNoName0516



**Operation Sword of
Justice**

Some Campaigns & Alliances in December 2024 of NoName57(16)

Recap

So...a lot happened in the 1,000 days of Russia's invasion of Ukraine...what do you message and when to prevent messaging fatigue?

- Uptick in scope and scale of operations
- Espionage, destruction, and disruption
- Shift to using edge devices to enable operational presence
- DDoS attacks on a daily basis
- #OpCountry operations 1-2 days
- Influence operations w/ DDoS attacks
- New alliances for integrated operations

When do you write an intelligence update?

- Is it a time of competition vs. time of conflict?

Relevance and Event Threshold?

- Impacted Industry
- Technical asset in our orgs environment
- ...and more?

What do we message?

- Impact potential
- Relation to previous known campaigns/capabilities
- Recommended mitigations/actions



Case Study 2: Iran-Hamas-Israel

It's All About the Proxies for Iran

Iranian hostility against Israel dates back to at least 1979 with overt tensions starting in the 1990s

- It started when the Supreme Leader disavowed the existence of Israel after taking control of the during the Iranian Revolution

Iran uses proxies to advanced regional objectives through traditional the use of proxies, including against Israel

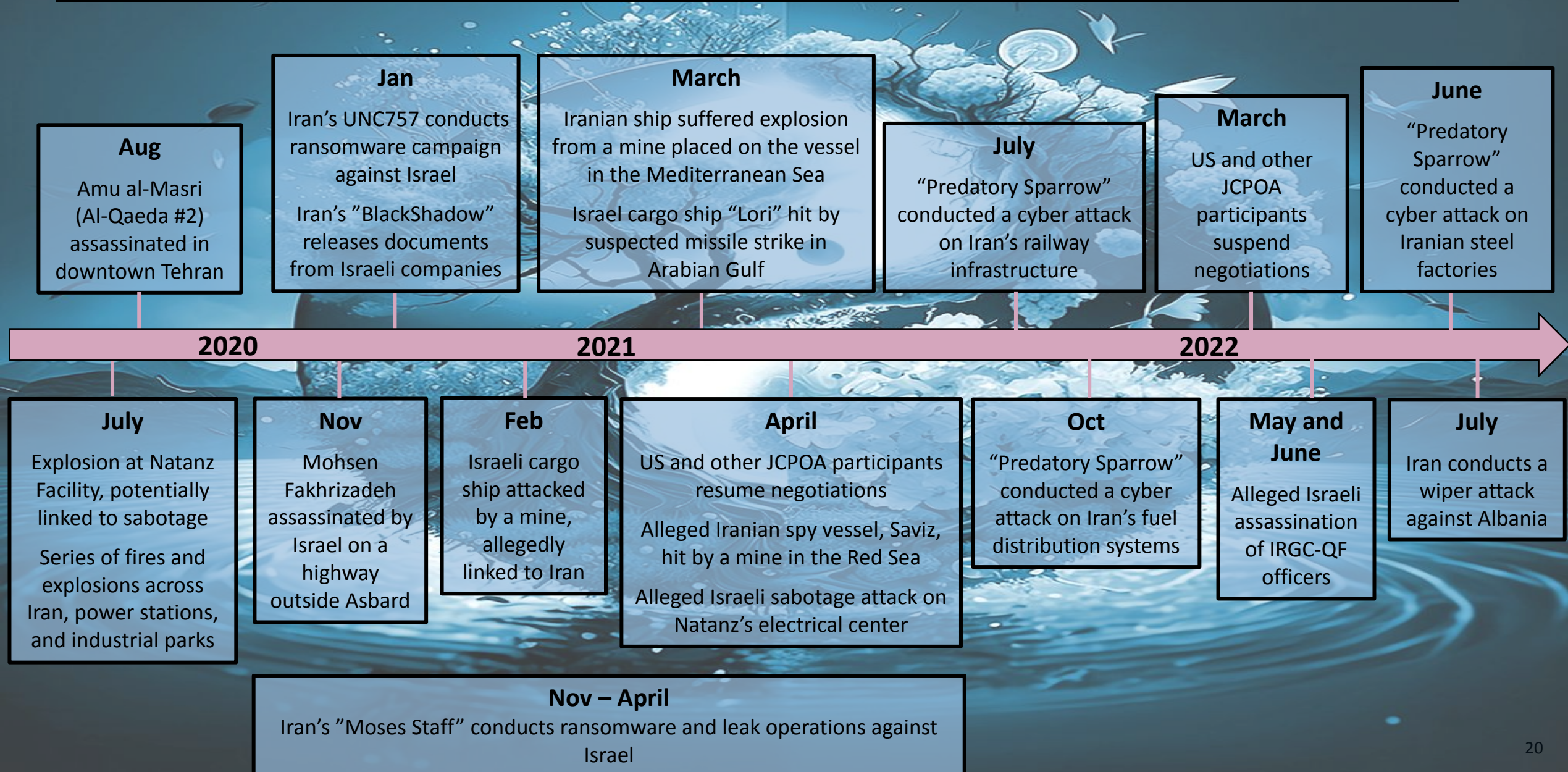
- Iran provides funding, military aid, advisement, and other support to its regional proxies (Hamas, Hezbollah, Shia, the Houthis, and the Assad regime)

Cyber support is independently and in coordination with non-cyber operations.

A “cyber war” has been taking place between Iran and Israel since at least 2020.

Date	Target/Incident	Impact	Perpetrator
24 April 2020	Israel Water Authority systems hacked to increase chlorine levels.	Water pumps at one station malfunctioned, but water supply not contaminated.	Western, Israeli officials attribute to Iran.
9 May 2020	Wiper at Shahid Rajaei Port.	Port operations interrupted for at least three days.	US and other foreign officials attribute to Israel.
20 Nov 2020	Insurer Shirbit database hacked w/ information on Israeli gov employees.	Data encrypted, stolen; hackers threatened to leak data but only did so partially.	BlackShadow hacker group, believed to be Iran.
9 July 2021	National railways come under cyber-attack.	Train service disrupted by fake cancellations and delays.	Unclear, CheckPoint attributes to regime opposition.
13 Oct 2021	Ransomware attack on Hillel Yaffe Medical Centre.	Administrative and medical systems brought down and locked.	BlackShadow hacker group, believed to be Iran.
26 Oct 2021	Petrol stations forced to close after smartcard services blocked.	Long lines formed at 4,300 stations; traffic disrupted. Digital billboards in Tehran and Isfahan were also compromised.	Iran blames Israel and the US for the attack.
30 Oct 2021	Database of Mor Medical Institute breached.	Medical records of 290,000 people posted on Telegram sites.	BlackShadow hacker group, believed to be Iran.
30 Oct 2021	Database of LGBTQ dating site Atraf breached.	Names, photos, and sexual preferences of thousands of users stolen; \$1 million USD ransom sought.	BlackShadow hacker group, believed to be Iran.

Cyber Attacks as Part of Iranian-Israeli Tensions, 2020-2022



Iranian Cyber Support to Hamas

Iran-based Cyber Actors Enabling Ransomware Attacks on US Organizations

Release Date: August 28, 2024

Alert Code: AA24-241A

Background on Threat Group and Prior Activity

This advisory outlines activity by a specific group of Iranian cyber actors that has conducted a high volume of computer network intrusion attempts against U.S. organizations since 2017 and as recently as August 2024. Compromised organizations include U.S.-based schools, municipal governments, financial institutions, and healthcare facilities. This group is known in the private sector by the names Pioneer Kitten, Fox Kitten, UNC757, Parasite, RUBIDIUM, and Lemon Sandstorm.[1][2] The actors also refer to themselves by the moniker Br0k3r, and as of 2024, they have been operating under the moniker “xplfinder” in their channels. FBI analysis and investigation indicate the group’s activity is consistent with a cyber actor with Iranian state-sponsorship.

The FBI previously observed these actors attempt to monetize their access to victim organizations on cyber marketplaces. A significant percentage of the group’s US-focused cyber activity is in furtherance of obtaining and maintaining technical access to victim networks to enable future ransomware attacks. The actors offer full domain control privileges, as well as domain admin credentials, to numerous networks worldwide. More recently, the FBI identified these actors collaborating directly with ransomware affiliates to enable encryption operations in exchange for a percentage of the ransom payments. These actors have collaborated with the ransomware affiliates NoEscape[3], Ransomhouse[4], and ALPHV (aka BlackCat) (#StopRansomware: ALPHV Blackcat). The Iranian cyber actors’ involvement in these ransomware attacks goes beyond providing access; they work closely with ransomware affiliates to lock victim networks and strategize on approaches to extort victims. The FBI assesses these actors do not disclose their Iran-based location to their ransomware affiliate contacts and are intentionally vague as to their nationality and origin.

Furthermore, the FBI has historically observed this actor conduct hack-and-leak campaigns, such as the late 2020 campaign known as Pay2Key.[5][6] The actors operated a .onion site (reachable through the Tor browser) hosted on cloud infrastructure registered to an organization previously compromised by the actors. (The actors created the server leveraging their prior access to this victim.) Following the compromise and the subsequent unauthorized acquisition of victim data, the actors publicized news of their compromise (including on social media), tagging accounts of victim and media organizations, and leaking victim data on their .onion site. While this technique has traditionally been used to influence victims to pay ransoms, the FBI does not believe the objective of Pay2Key was to obtain ransom payments. Rather, the FBI assesses Pay2Key was an information operation aimed at undermining the security of Israel-based cyber infrastructure.

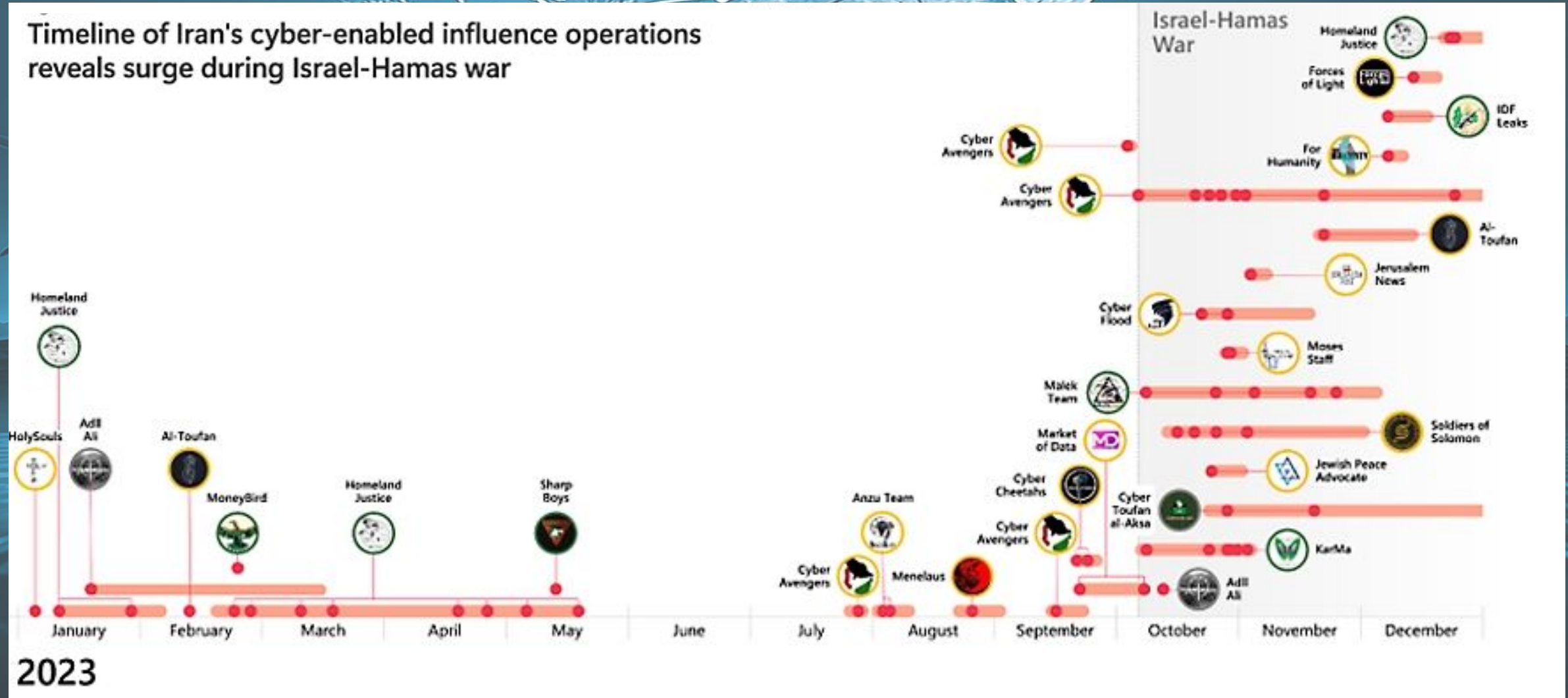
Attribution Details

FBI investigation identified that the Iranian cyber actors conduct malicious cyber activity, which FBI assessed to be in support of the GOI. The FBI judges this activity to be separate from the previously referenced ransomware-enabling activity. This group directs their activity towards countries and organizations consistent with Iranian state interests, and typically not of interest to the group’s ransomware affiliate contacts, such as U.S. defense sector networks, and those in Israel, Azerbaijan, United Arab Emirates. Instead, it is intended to steal sensitive information from these networks, suggesting the group maintains an association with the GOI. However, the group’s ransomware activities are likely not sanctioned by the GOI, as the actors have expressed concern for government monitoring of cryptocurrency movement associated with their malicious activity.

The group uses the Iranian company name Danesh Novin Sahand (identification number 14007585836), likely as a cover IT entity for the group’s malicious cyber activities.

Iranian Cyber Support to Hamas

Timeline of Iran's cyber-enabled influence operations reveals surge during Israel-Hamas war



Iranian Cyber Support to Hamas

Iran has a long standing history of using disruptive or destructive cyber operations to respond to actual or perceived aggressions. Post 7 Oct 2023 efforts were no different...

- According to Microsoft, “As the Israel-Hamas war broke out ... Iran surged support to Hamas ... combining targeted hacks with influence operations...”
- ...By late October, nearly all of Iran’s influence and major cyber actors focused on Israel ... in an “all-hands-on-deck” campaign against Israel.
- ...Unlike previous Iranian cyber attacks, all of its destructive cyber operations against Israel in this war were complemented with online influence operations.”



In the Name of God The Merciful 2023-10-30

We (Cyber Av3ngers) have infiltrated to 10 Water treatment stations located in Hedera, Palmachim, Sorek, Ashkelon, Haifa, Chorazim, Kfar Haruv, Taberiye, Eilat and Daniyal.

Once we obtained access to their network, Established to manipulate, wipe and destruct all industrial equipment such as SCADA systems, PLCs, sensors and HMIs.

To the residents of the occupied territories:

Your and your prisoners of war life
is worthless to your rulers!

 Cyber Av3ngers

IRGC-Affiliated Cyber Actors Exploit PLCs in Multiple Sectors, Including US Water and Wastewater Systems Facilities

Last Revised: December 18, 2024

Alert Code: AA23-335A

Overview and History of Threat Actor Activity

CyberAv3ngers is an Iranian IRGC-affiliated cyber persona (also known as CyberAveng3rs or Cyber Avengers) that has claimed responsibility for numerous attacks against critical infrastructure organizations primarily in the water and energy sectors in the United States, Israel, and other countries.^{[1][2][3][4][5][6][7][8][9][10]} Since 2020, CyberAv3ngers has claimed responsibility for cyberattacks in Israel, although several of these claimed compromises of critical infrastructure organizations in Israel are false.^[3] In October 2023, CyberAv3ngers claimed credit on their Telegram Channel for cyberattacks against Israel-based PLCs. The PLCs were internet-facing, used Unitronics’ default passwords or no password, and connected to default ports—vulnerabilities that were likely exploited by the actors. CyberAv3ngers also reportedly has connections to another IRGC-linked group known as Soldiers of Solomon. The observed activity includes the following:

- Between Sept. 13, 2023, and Oct. 30, 2023, the CyberAv3ngers Telegram channel displayed both legitimate and false claims of multiple cyberattacks against Israel. CyberAv3ngers targeted Israeli PLCs in the water, energy, shipping, and distribution sectors.
- Beginning on Nov. 22, 2023, IRGC cyber actors accessed multiple U.S.-based WWS facilities that operate HMI-capable Unitronics Vision Series PLCs likely by compromising internet accessible devices with default or no passwords. The targeted PLCs displayed the defacement message, “You have been hacked, down with Israel. Every equipment ‘made in Israel’ is Cyberav3ngers legal target.”

(Update Dec. 18, 2024)

Threat Actor Activity Against U.S.-Based Unitronics Devices

Between November 2023 and January 2024, CyberAv3ngers targeted U.S.-based Unitronics PLC devices used in multiple critical infrastructure industries, including the WWS Sector, likely in four separate waves of cyberattacks. The actors compromised at least 75 devices, including at least 34 in the WWS Sector in the United States.



Case Study 3: China-Taiwan

Historical Context and Significance

Taiwan is one of China's "Five Poisons".

- Tensions date back to 1931 with an 18-year civil war between the Chinese Communist Party and the Nationalist Party.
- In December 1949, 2 million individuals from the Chinese Nationalist Party (Kuomintang) retreated from China to Taiwan. To this day, each claims ownership over the other.
 - The Chinese Nationalist Party called Taiwan the "Republic of China" (ROC), which in response, prompted the Chinese Communist Party (CCP) to call the mainland the "Peoples' Republic of China" (PRC).
- No armistice treaty signed at the end of the Chinese civil war. "Cross-strait relations" is the mutually agreed upon, neutral term by both parties.
- During the 1990s Taiwan elected to advocate to operate as a singular country, moving away from the one-party authoritarian state that had previously committed to eventual unification with China.
 - In response, the PLA attempted to influence the outcome of 1996 Taiwan election, leading to the "3rd Taiwan Strait crisis".
 - The anti-secession law passed by the Chinese National People's Congress in 2005 was a significant escalation point. This essentially reserved the right for China to invade Taiwan should Taiwan attempt to pursue its independence.



Cyber Operations Against Taiwan...

- The Taidoor malware used since at least 2008 is probably the most well-known.
- Second to what took place during and the lead up to surprise visit to Taiwan from U.S. House of Representatives Speaker Nancy Pelosi's amidst rising tensions with China in August 2022...
 - Several Taiwanese government websites were taken offline in a DDoS attack at the rate of 23 times the size of normal traffic loads, according to Taiwan's Digital Minister.
 - The affected websites included the Presidential website, the Ministry of National Defense, the Ministry of Foreign Affairs, and the website of Taiwan Taoyuan International Airport, where Pelosi's plane landed.
 - In addition, 7-11 stores, amongst others, found their TV systems compromised and subsequently defaced, with anti-Pelosi messaging suggesting she leave Taiwan immediately. This took place concurrent to China initiating a military exercise in the region in advance of and during Pelosi's visit to signal displeasure and as a vague threat that it could attack the island at any point in time with its Naval fleet.

Chinese Threat Groups Targeting Taiwan

APT1

APT2

APT5

APT12

APT16

APT17

APT22

APT24

APT41

Conference Crew

...and the Broader Backdrop

- Enactment of laws to potentially amass an arsenal of zero days from Chinese cyber security companies
- Seemingly centralized administration of ORB networks by Chinese companies
- Re-org from the SSF to the ISF
- “Pre-positioning” on critical infrastructure

PRC State-Sponsored Actors Compromise and Maintain Persistent Access to U.S. Critical Infrastructure

Release Date: February 07, 2024

Alert Code: AA24-038A

The Cybersecurity and Infrastructure Security Agency (CISA), National Security Agency (NSA), and Federal Bureau of Investigation (FBI) assess that People's Republic of China (PRC) state-sponsored cyber actors are seeking to pre-position themselves on IT networks for disruptive or destructive cyberattacks against U.S. critical infrastructure in the event of a major crisis or conflict with the United States.

CISA, NSA, FBI and the following partners are releasing this advisory to warn critical infrastructure organizations about this assessment, which is based on observations from the U.S. authoring agencies' incident response activities at critical infrastructure organizations compromised by the PRC state-sponsored cyber group known as Volt Typhoon (also known as Vanguard Panda, BRONZE SILHOUETTE, Dev-0391, UNC3236, Voltzite, and Insidious Taurus):

- U.S. Department of Energy (DOE)
- U.S. Environmental Protection Agency (EPA)
- U.S. Transportation Security Administration (TSA)
- Australian Signals Directorate's (ASD's) Australian Cyber Security Centre (ACSC)
- Canadian Centre for Cyber Security (CCCS), a part of the Communications Security Establishment (CSE)
- United Kingdom National Cyber Security Centre (NCSC-UK)
- New Zealand National Cyber Security Centre (NCSC-NZ)

The U.S. authoring agencies have confirmed that Volt Typhoon has compromised the IT environments of multiple critical infrastructure organizations—primarily in [Communications](#), [Energy](#), [Transportation Systems](#), and [Water and Wastewater Systems](#) Sectors—in the continental and non-continental United States and its territories, including [Guam](#). Volt Typhoon's choice of targets and pattern of behavior is not consistent with traditional cyber espionage or intelligence gathering operations, and the U.S. authoring agencies assess with high confidence that Volt Typhoon actors are pre-positioning themselves on IT networks to enable lateral movement to OT assets to disrupt functions. The U.S. authoring agencies are concerned about the potential for these actors to use their network access for disruptive effects in the event of potential geopolitical tensions and/or military conflicts. CCCS assesses that the direct threat to Canada's critical infrastructure from PRC state-sponsored actors is likely lower than that to U.S. infrastructure, but should U.S. infrastructure be disrupted, Canada would likely be affected as well, due to cross-border integration. ASD's ACSC and NCSC-NZ assess Australian and New Zealand critical infrastructure, respectively, could be vulnerable to similar activity from PRC state-sponsored actors.

What and When to Warn

As Tensions Rise	During a Flashpoint
CTI Role: Strategic warning w/ BIA + Prioritized list of threat actors of interest	CTI Role: Hunt and IR support w/ playbook development
Relevance and Likelihood? ● Direct impact to business operations? ○ Do we operate in Taiwan or China? ● High risk industry? e.g. Telecommunications, Defense, High Tech, National Gov, Media, or Energy. ● Military or intelligence service contractors or supplier? ● Data transit dependencies? ● Risk level increase for espionage or attack/disruption pre-positioning? ● IO/brand reputation hard to gauge, but keep tabs on public discourse from senior execs commenting on the evolving situation.	????

Questions?



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